

aNLP | Wordsmiths

# DiffusEdit

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# Overview

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# Abstract

Reference articles go out of date as events develop, and updating them by hand is slow. Using a language model to automate the update rewrites text that was already correct and leaves no record of what changed. DiffusEdit instead edits the article in place. The frozen model rewrites only the stale positions, and the system repeats the process on sentences that the rewrites themselves made wrong.



# Problem

- **Reference text ages rapidly:** Real-world events (news, Wikipedia) constantly makes existing facts out-dated.
- **Current LLM solutions are brute-force:** Auto-regressive generation rewrites the entire article from scratch, which is slow and blindly overwrites perfectly correct sentences.
- **The "Cascading Edit" Dilemma:** Facts are logically connected. If you change a fact in paragraph one, sentences in paragraph three might suddenly become completely unsupported.

## Example:

**Sentence 1:** “Three people died”

**Sentence 2:** “The small death toll limited the political fallout”

**A week later, let’s say the death toll increases to 42.**

**S1:** wrong

**S2 :** wrong by dependence



# Contributions

- A **sentence-level KEEP/EDIT/DROP classifier** over pooled hidden states, which restricts token scoring to EDIT sentences and adds deletion through DROP.
- An iterative edit loop that returns broken sentences to the pipeline, comparing three **cascade detectors**: mention overlap, staleness delta, and a learned consequence head.
- **Variable-length repair through multi-length infilling** and reranking for fluency and evidence support.
- **Sentence- and token-level staleness labels** derived from the dataset by alignment, plus a mined and hand-checked set of cascade pairs.
- An **evaluation** separating detection quality, update quality, minimality, and cascade recall, with baselines, oracles, and ablations.

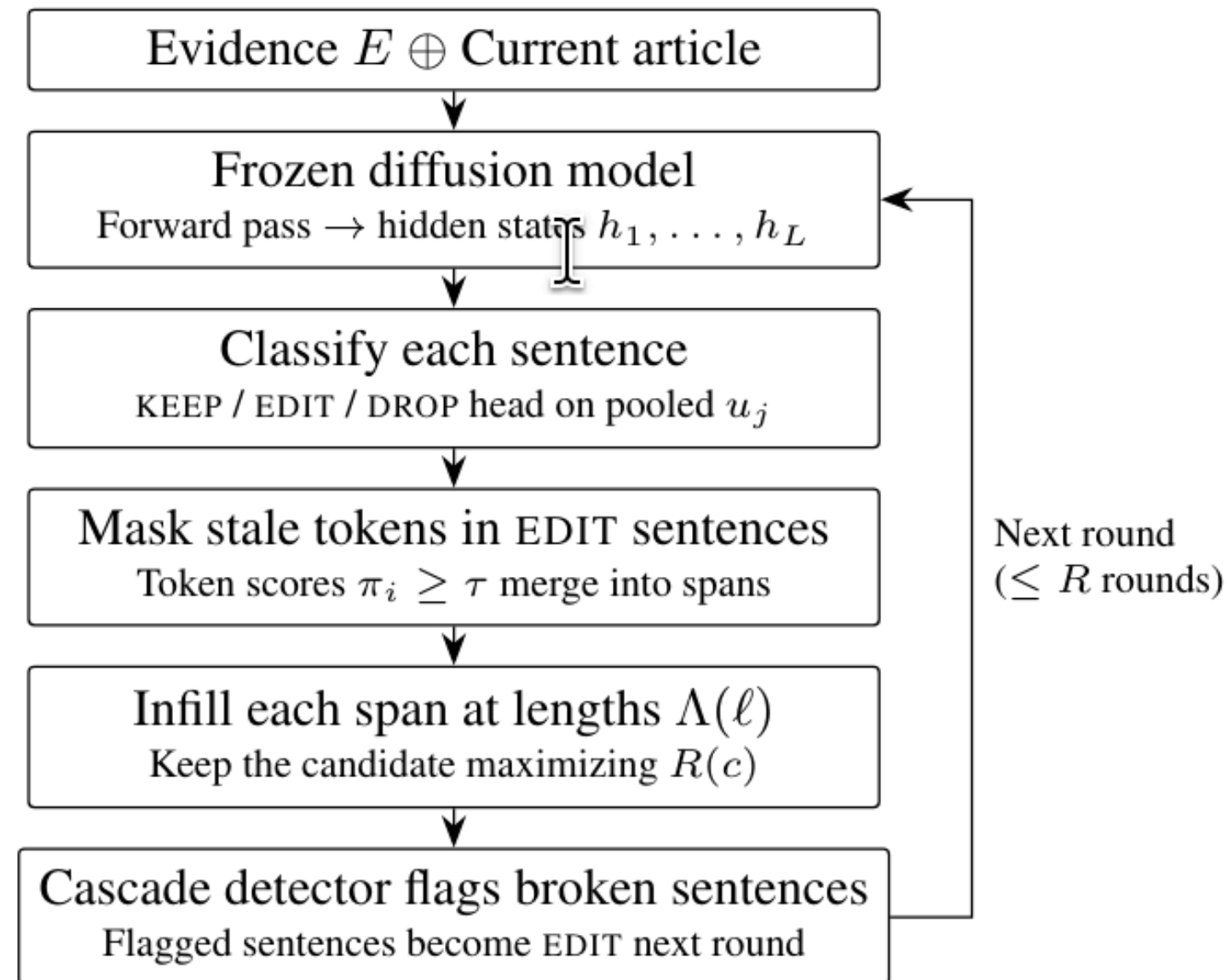


# Dataset

- To train our classifiers, we used the FRUIT dataset, which provides paired Wikipedia snapshots (outdated vs. updated) alongside the new evidence.
- Because FRUIT did not have manual, sentence-level annotations, we used text-alignment heuristics to automatically label sentences as KEEP, EDIT, or DROP.
- However, this method led to a HUGE problem: Wikipedia editors not only update facts but also concurrently do stylistic rewrites such as switching from active to passive voice. Our algorithmic heuristic blindly flags these as factual EDITs.
- Our frozen diffusion backbone (LLaDA-8B) encodes text based on the semantics of the facts rather than just surface-level grammar and syntax.
- A stylistic rewrite has the exact same hidden-state vector as a factually clean KEEP sentence.
- This semantic collision causes 13% label noise in edit labels across the dataset. This diminishes the theoretical upper bound of our linear classifiers.



# Pipeline



# Current Progress

**Feature Extraction**

**Masking and Variable  
Length Infilling**

**Token Staleness  
Identification**

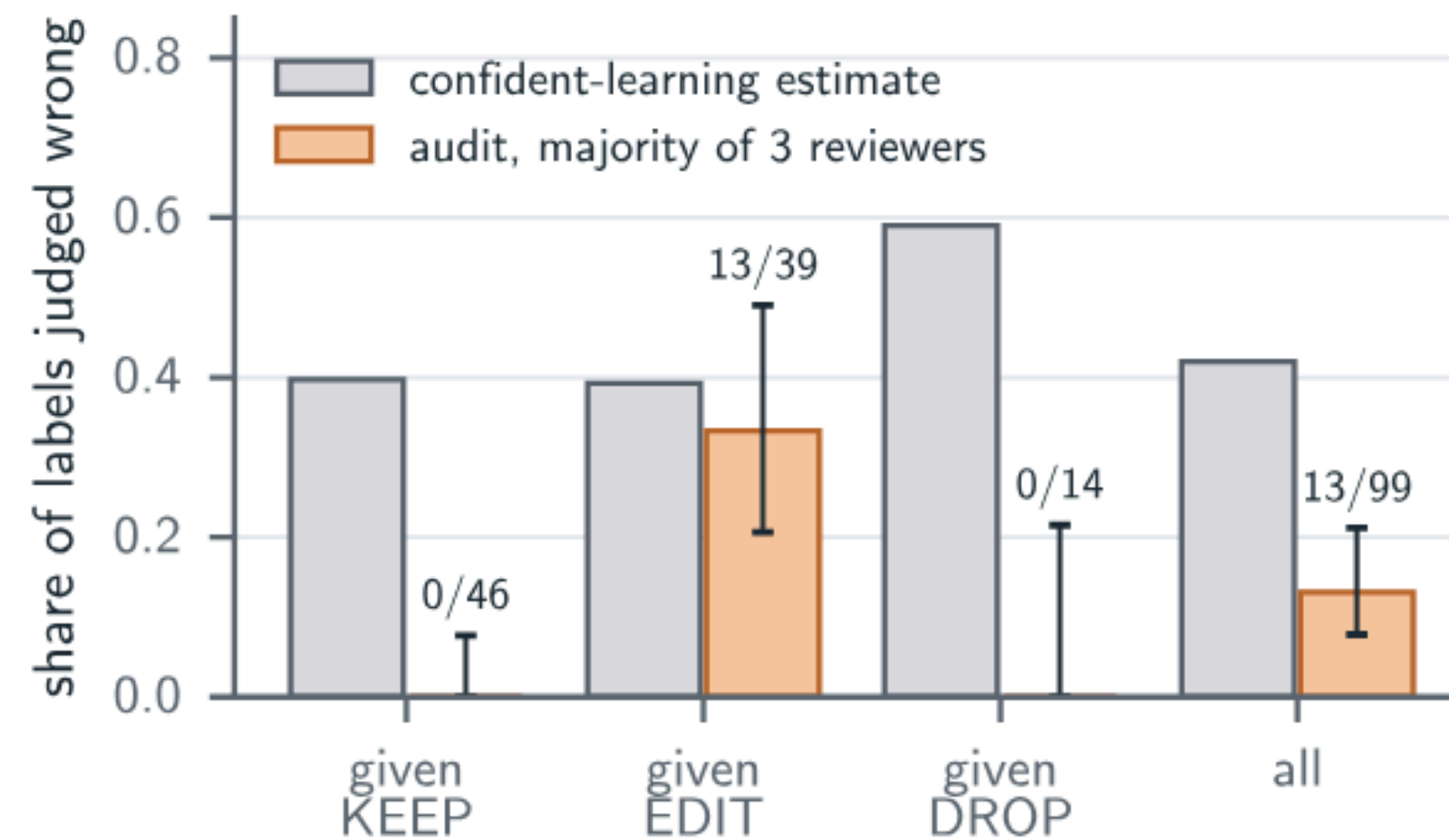
**Sentence-Level  
Classifier**

**Ablations:**

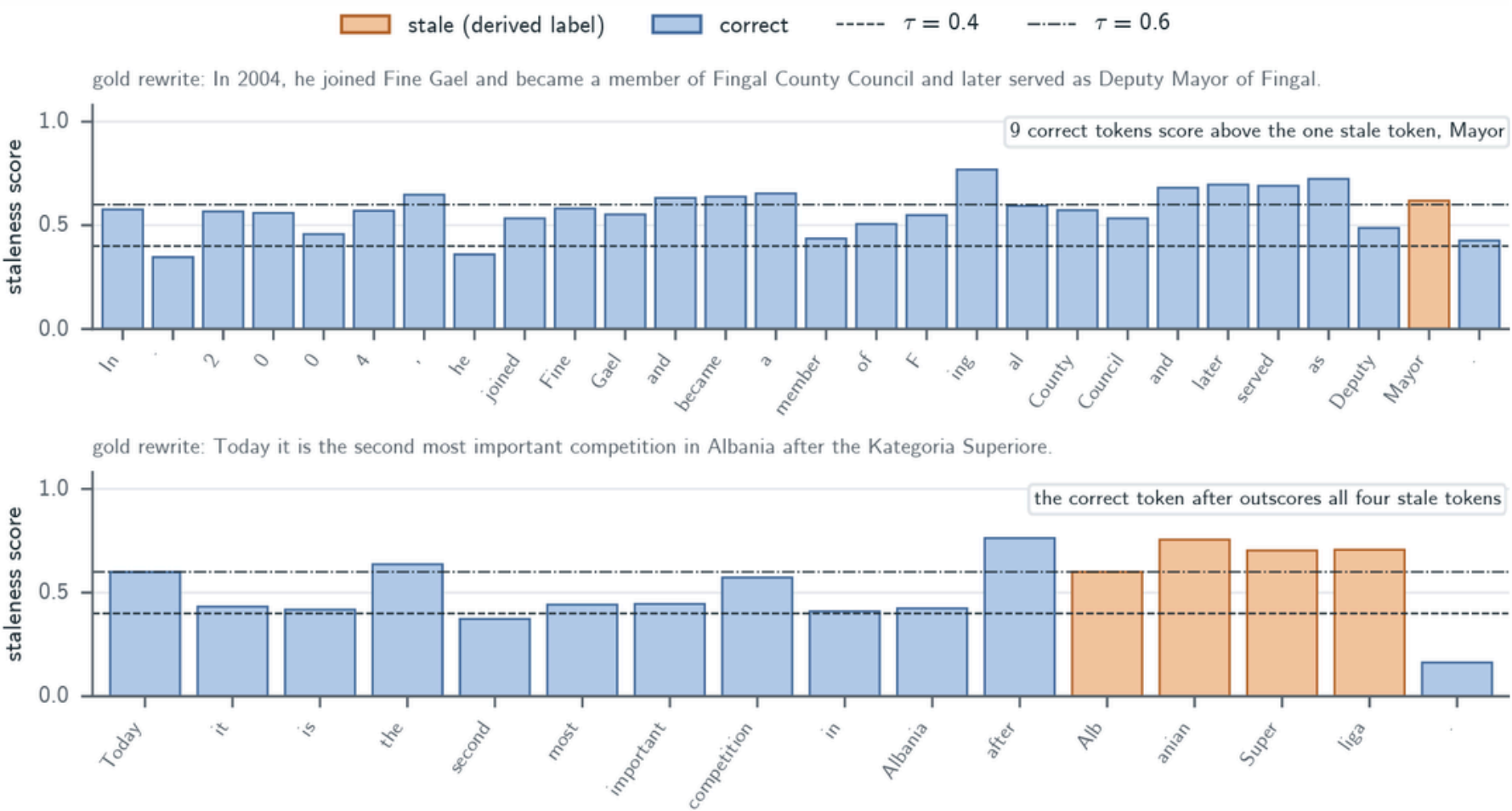
- Non-diffusion baselines
- Sentence-Level Classifier with evidence



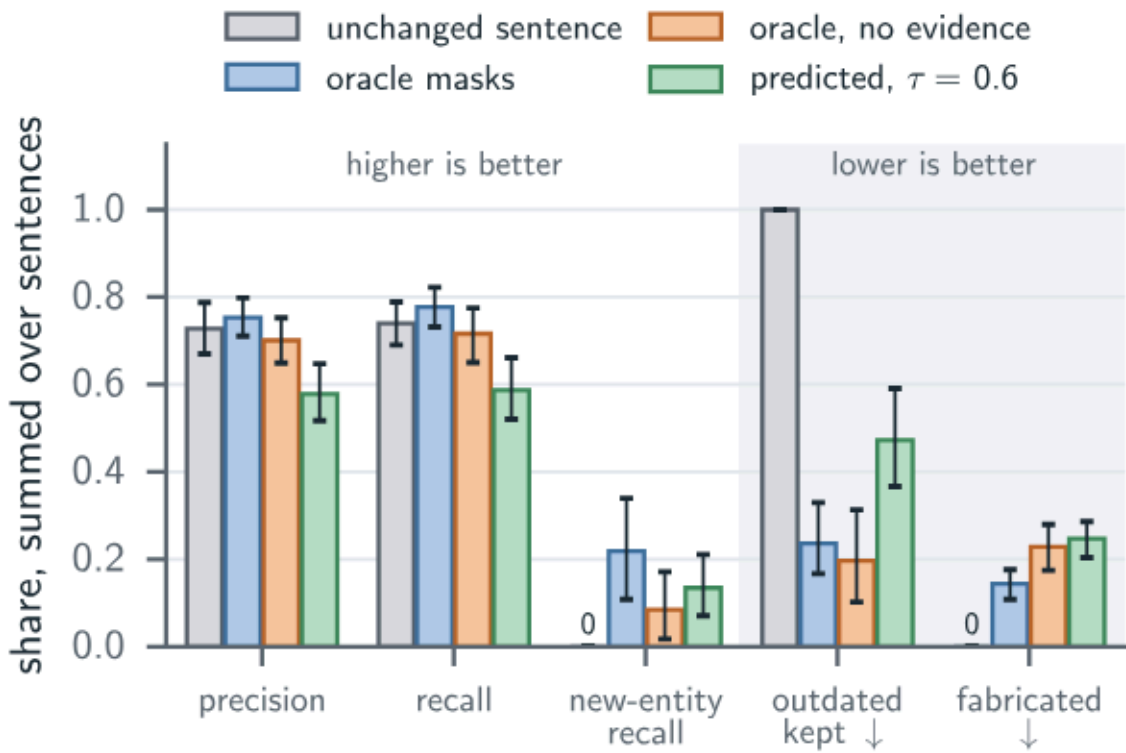
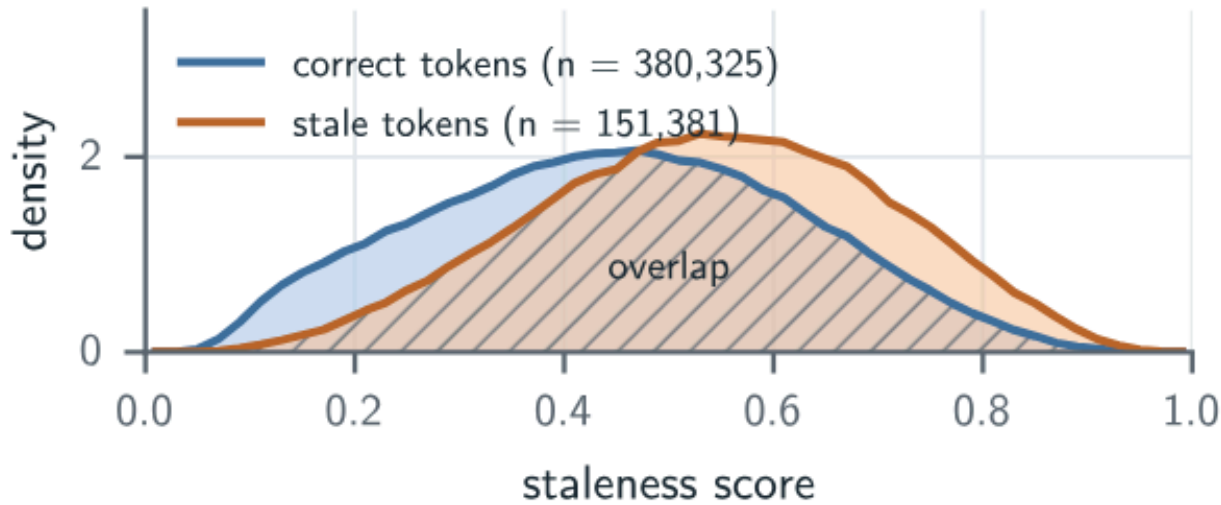
# Results



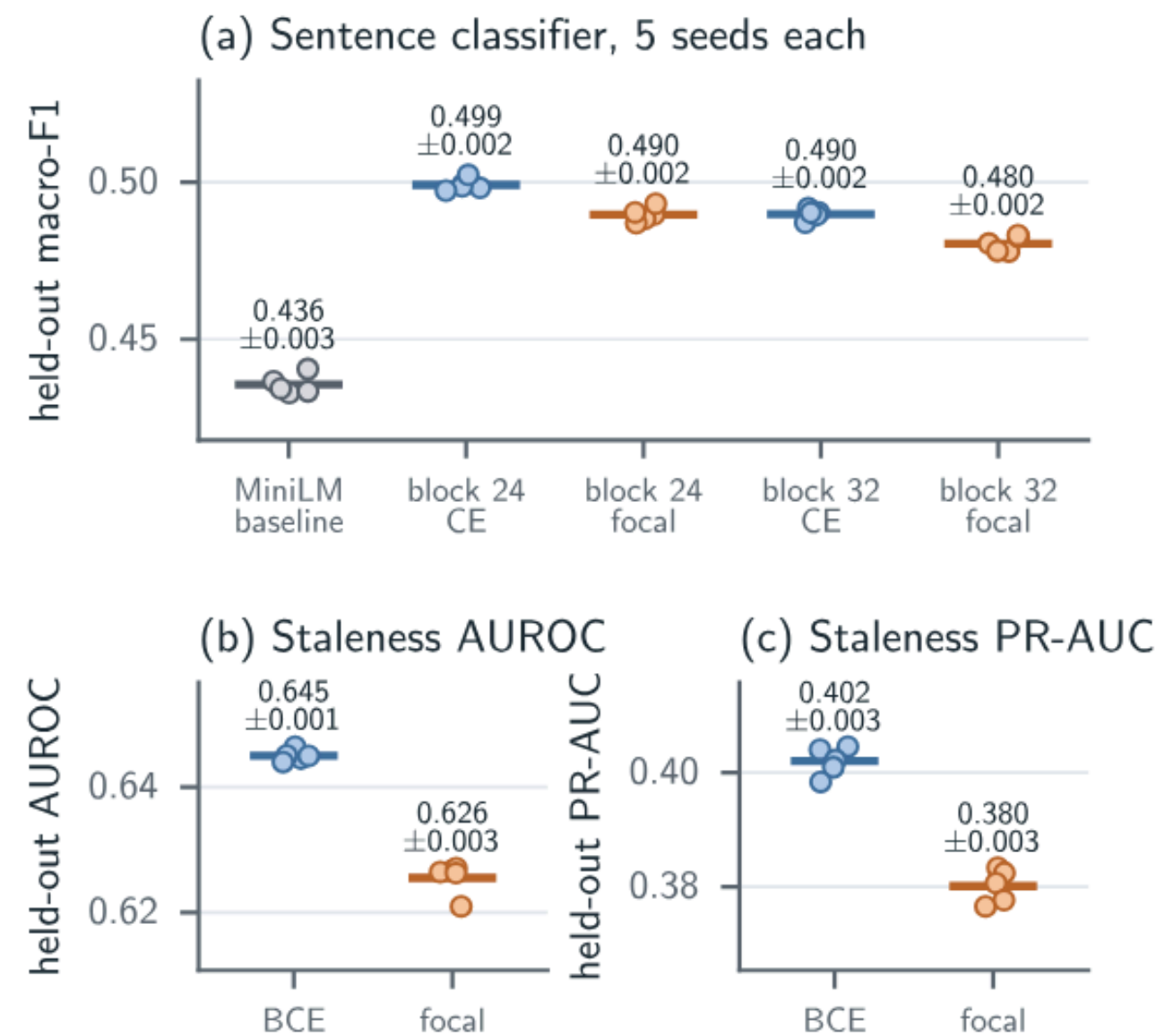
# Results



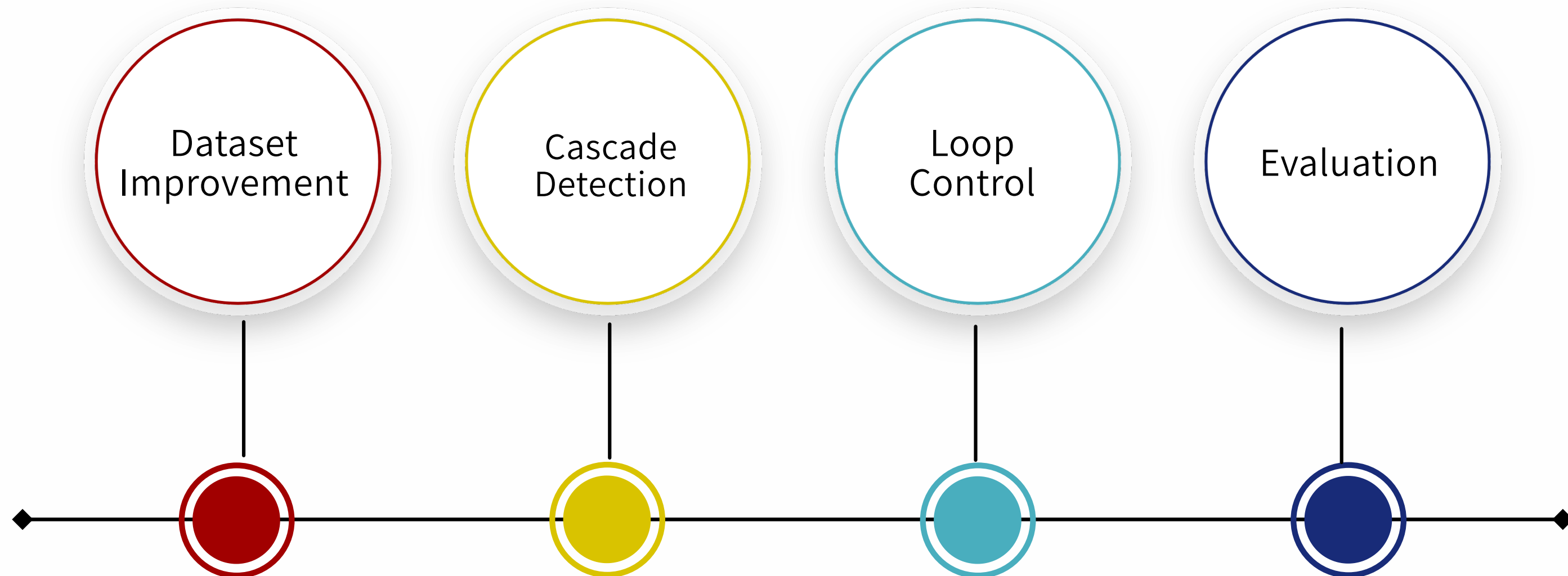
Scores by label (AUROC 0.645)



# Results



# Next Steps



# References

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# Questions?

Your doubt is [MASK] - we'll infill it



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# THANK YOU

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